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Inventor(s)	Xu; Xinglong et al.

Ball bearing slide

Abstract

A ball bearing slide (**100**) includes: at least one outer slide (**1**), an inner slide (**2**) located within the outer slide, a ball cage (**4**) located between the inner slide and the outer slide, and three rows of balls (**3**) rollably arranged in the ball cage; wherein the three rows of the balls (**3**) includes a first ball row (**31**), a second ball row (**32**), and a third ball row (**33**) located between the first and second ball rows in a vertical direction; wherein the inner slide includes a first curved part (**24**) against the first ball row, a second curved part (**25**) against the second ball row, and a third curved part (**26**) against the third ball row; the third curved part is located between the first ball row and the second ball row in the vertical direction, and is symmetrically arranged relative to a symmetry axis of a line connecting centers of the first ball row and the second ball row. With the first, second and third curved parts, internal space of the ball bearing slide for containing the inner slide is increased, which rationalizes the ball cage arrangement and optimizes the ball bearing slide design.

Inventors:	**Xu; Xinglong (Wuxi, Jiangsu, CN), Zhang; Haiwen (Wuxi, Jiangsu, CN), Zou; Lian (Wuxi, Jiangsu, CN), Qian; Feng (Wuxi, Jiangsu, CN), Dai; Kai (Wuxi, Jiangsu, CN), Ji; Qiang (Wuxi, Jiangsu, CN)**
Applicant:	**WUXI HAIDAER PRECISION SLIDES CO., LTD** (Wuxi, Jiangsu, CN)

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Background/Summary

BACKGROUND OF THE PRESENT INVENTION

Field of Invention

[0001] The present invention relates to a ball bearing slide, and more particularly to a hidden ball bearing slide that can adapt to a small installation space.

Description of Related Arts

[0002] In a conventional three-row ball bearing slide, such as the pulling guide device disclosed in Chinese patent CN103096756B, the inner slide is placed in at least one outer slide and is supported by three rows of supporting balls which are arranged in parallel along the length direction. The three rows of supporting balls are restricted within the length of the outer slide by ball cages. In a cross section of the outer slide, the three rows of supporting balls are positioned at three adjacent inner corners of the outer slide by the ball cage, so that the ball centers of the three rows of supporting balls are distributed as a right-angle triangle. Usually, the three rows of supporting balls can steadily support the inner slide in the outer slide, so as to effectively reduce the wobble gap of the inner slide, and prevent the inner slide from radial wobble in the cross section of the outer slide. Therefore, the conventional three-row ball bearing slides are mostly used in large capacity drawers or pull-out baskets of furniture and household appliances such as refrigerators, ovens, dishwashers, and other furniture to ensure the stability during opening or closing the drawer or basket. However, in practice, bearing capacity the conventional three-row ball bearing slide may be insufficient. When the drawer or basket is too long or heavily loaded, especially when the outer slide bears a large pressure or tension which is perpendicular to the outer slide plane, the curved surfaces of the inner slide that are contacted and supported by the three rows of supporting balls can easily be deformed due to uneven force. Under serious cases, this will lead to jamming of the drawer or basket. Furthermore, the conventional three-row ball bearing slide has a large size, and thus cannot adapt to the small space for hidden slide installation.

[0003] Therefore, it is really necessary to provide a novel ball bearing slide to overcome the above defects.

SUMMARY OF THE PRESENT INVENTION

[0004] An object of the present invention is to provide a ball bearing slide that adapts to a small space for hidden slide installation.

[0005] Accordingly, in order to accomplish the above object, the present invention provides a ball bearing slide, comprising: at least one outer slide, an inner slide located within the outer slide, a ball cage located between the inner slide and the outer slide, and three rows of balls rollably arranged in the ball cage; wherein the inner slide is supported by the balls to slide relative to the outer slide along an arrangement direction of the three rows of the balls; the outer slide comprises a bottom wall, as well as a long wall portion and a short wall portion which are formed by integrally extending both sides of the bottom wall and are arranged along a length direction of the bottom wall; the three rows of the balls comprises a first ball row and a second ball row contacting an interior side of the long wall portion, and a third ball row contacting an interior side of the short wall portion and located between the first ball row and the second ball row in a vertical direction; wherein the inner slide comprises a first curved part against a curved surface of the first ball row facing the inner slide, a second curved part against a curved surface of the second ball row facing the inner slide, and a third curved part against a curved surface of the third ball row facing the inner slide; the third curved part is located between the first ball row and the second ball row in the vertical direction, and is symmetrically arranged relative to a symmetry axis of a line connecting

centers of the first ball row and the second ball row.
[0006] Preferably, the third curved part has a bottom side in a width direction perpendicular to the vertical direction; the inner slide has a central axis C-C extending in the vertical direction and passing through a central point of the inner slide; and the bottom side exceeds the central axis C-C in the width direction.
[0007] Preferably, the ball cage comprises a fourth holding portion corresponding to the short wall portion for holding the third ball row, wherein the fourth holding portion extends flatly along the vertical direction.
[0008] Preferably, the ball cage further comprises a first holding portion for holding the first ball row, a third holding portion for holding the second ball row, and a first connecting portion connecting the first holding portion and the third holding portion; wherein the first connecting portion extends flatly along the vertical direction, and a portion of the long wall portion corresponding to the first connecting portion also extends flatly along the vertical direction.
[0009] Preferably, the outer slide further comprises: a first support formed at an end of the long wall portion and having a curved internal contour, a second support formed at an end of the short wall portion and having a curved internal contour, a first corner corresponding to the first support and vertically arranged at a top of the long wall portion, a second corner corresponding to the second support and vertically arranged at a top of the short wall portion, and a connecting wall connecting the first corner and the second corner; wherein the first holding portion of the ball cage is arranged corresponding to the first corner, a second holding portion of the ball cage is arranged corresponding to the second corner, the third holding portion is arranged corresponding to the first support, and the fourth holding portion is arranged corresponding to the second support; the first ball row is restricted by the first holding portion corresponding to the first corner portion, the second ball row is restricted by the third holding part corresponding to the first support, and the third ball row is restricted by the fourth holding part corresponding to the second support.
[0010] Preferably, an angle of an internal contour of the third curved part is 100°-110° for holding the third ball row.
[0011] Preferably, the inner slide further comprises a second convex part connecting the second curved part and the third curved part, wherein the second convex part is formed by folding and bending a portion between the first curved part and the third curved part and rolling the part at an angle towards the second holding portion; the second convex part is located between the first ball row and the third ball row.
[0012] Preferably, a thickness of the second convex part equals a distance between the first ball row and the third ball row, a dimension of the second convex part is smaller than a total thickness of two rolled raw materials of the inner slide.
[0013] Preferably, the second convex part is spaced from the second holding portion with a second spacing therebetween; one side of the second convex part, which faces the second holding portion, is rolled into a flat surface to place the ball cage.
[0014] Preferably, the inner slide further comprises a first convex part connecting the first curved part and the second curved part, which corresponds to a position of the third curved part, wherein the bottom side of the third curved part is pressed inwardly towards the first convex part; the first convex part is formed by folding and bending a portion between the first curved portion and the second curved portion.
[0015] Preferably, the first convex part is located between the first ball row and the second ball row; the first convex part is spaced from the first connecting part in a width direction with a first spacing therebetween; one side of the first convex part, which faces the first connecting portion, is rolled into a flat surface to place the ball cage.
[0016] Preferably, a tail portion of the third curved part is cocked towards the third ball row; a tangent line of the tail portion of the third curved part and a tangent line of the short wall portion together form an opening with a first angle; an end of the fourth holding portion extends out

through the opening.
[0017] According to the ball bearing slide of the present invention, the first curved part, the second curved part and the third curved part are designed on the inner slide, the first convex part is located between the first curved part and the second curved part, and the second convex part is located between the second curved part and the third curved part. As a result, the third curved part presses towards the interior side to exceed the other side of the central axis of the ball bearing slide. At the same time, by rolling the first convex part and the second convex part, a sufficient internal space is generally provided in the ball cage, so as to rationalize the arrangement of the ball cage and improve the overall bearing capacity of the ball bearing slide.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0018] FIG. **1** is a perspective view of a ball bearing slide of the present invention;
[0019] FIG. **2** is viewed from another direction of FIG. **1**;
[0020] FIG. **3** is an exploded view of the ball bearing slide of the present invention;
[0021] FIG. **4** is viewed from another direction of FIG. **3**;
[0022] FIG. **5** is a partially exploded view of the ball bearing slide of the present invention;
[0023] FIG. **6** is a front elevational view of the ball bearing slide of the present invention;
[0024] FIG. **7** is a cross-sectional view of the ball bearing slide of the present invention;
[0025] FIG. **8** is a partially enlarged view of FIG. **7**; and
[0026] FIG. **9** is another partially enlarged view of FIG. **7**.
DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT
[0027] In the description of the present invention, it should be noted that the terms “upper”, “lower”, “inner”, “outer”, etc. indicate orientation or positional relationships that are based on the orientation or positional relationships shown in the accompanying drawings and are intended only to facilitate and simplify the description of the present invention. Such terms are not intended to indicate or imply that the device or component referred to must have a particular orientation or be constructed and operated in a particular orientation. Therefore, such terms cannot be construed as limiting the present invention.
[0028] In the description of the present invention, it should be noted that unless otherwise expressly specified and limited, the term “connection” should be understood in a broad sense. To persons of ordinary skill in the art, the specific meaning of the above terms in the context of the present invention can be understood according to specific situations.
[0029] Hereinafter, the ball bearing slide **100** of the present invention will be described in conjunction with FIGS. **1-9**.
[0030] A ball bearing slide **100** is provided, comprising: at least one outer slide **1**, an inner slide **2** located within the outer slide **1**, a ball cage **4** located between the inner slide **2** and the outer slide **1**, and three rows of balls **3** rollably arranged in the ball cage **4**. According to the embodiment, there are two outer slides **1**, namely a first outer slide **11** and a second outer slide **12** symmetric to the first outer slide **11**. The first outer slide **11** is fixed to a cabinet of furniture or home appliance by a fixing bracket **5**, and the second outer slide **12** is fixed to an external wall of a drawer. The inner slide **2** is supported in the outer slide **1** by the three rows of the balls **3** set in parallel along a length direction, and the three rows of the balls **3** are limited by the ball cage **4** within a length of the outer slide **1**. The inner slide **2** is supported by the balls **3** to translate along the outer slide **1**. The ball cage **4** has three grooves matching the three rows of the balls, and one groove restricts one row of the balls **3**. Slots **48** are spaced on each of the three grooves for holding and restricting the balls **3**.
[0031] Referring to FIGS. **1-9**, the first outer slide **11** and the second outer slide **12** have the same structure, and the second outer slide **12** will be illustrated as an example.

[0032] The outer slide **1** comprises a bottom wall **13** fixed with the fixing bracket **5**, as well as a long wall portion **14** and a short wall portion **15** which are formed by integrally extending both sides of the bottom wall **13** and are arranged along a length direction of the bottom wall **13**. The outer slide **1** further comprises: a first support **141** formed at an end of the long wall portion **14** and having a curved internal contour, a second support **151** formed at an end of the short wall portion **15** and having a curved internal contour, a first corner **16** corresponding to the first support **141** and vertically arranged at a top of the long wall portion **14**, a second corner **17** corresponding to the second support **151** and vertically arranged at a top of the short wall portion **15**, and a connecting wall **18** connecting the first corner **16** and the second corner **17**. An end of the first support **141** is defined as a first end **142**, and an end of the second support **151** is defined as a second end **152**.
[0033] According to the embodiment, there are two ball cages **4**, one ball cage **4** corresponds to one outer slide **1**. The ball cage **4** is positioned relative to the inner slide **2**. If the ball bearing slide **100** is applied in heat resistant or harsh environments such as microwave oven, the ball cage **4** is made of steel material; and if the ball bearing slide **100** is applied in environments such as drawer furniture, the ball cage **4** is made of plastic material. The ball cage **4** comprises a first holding portion **41** for holding the first ball row **31**, a second holding portion **42** corresponding to the second corner **17**, a third holding portion **43** for holding the second ball row **32**, and a fourth holding portion **44** corresponding to the short wall portion **15** for holding the third ball row **33**, wherein the fourth holding portion **44** extends flatly along the vertical direction. The first holding portion **41** is arranged corresponding to the first corner **16**, the third holding portion **43** is arranged corresponding to the first support **141**, and the fourth holding portion **44** is arranged corresponding to the second support **151**. The first ball row **31** is restricted by the first holding portion **41** corresponding to the first corner portion **16**, the second ball row **32** is restricted by the third holding part **43** corresponding to the first support **141**, and the third ball row **33** is restricted by the fourth holding part **44** corresponding to the second support **151**. The ball cage **4** further comprises a first connecting portion **45** arranged in the vertical direction and connecting the first holding portion **41** and the third holding portion **43**, a second connecting portion **46** arranged in a width direction of the ball bearing slide **100** and connecting the first holding portion **41** and the second holding portion **42**, and a third connecting portion **47** arranged in the vertical direction and connecting the second holding portion **42** and the fourth holding portion **44**. The first connecting portion **45** extends flatly along the vertical direction, and a portion of the long wall portion **14** corresponding to the first connecting portion **45** also extends flatly along the vertical direction. The first connecting portion **45**, the second connecting portion **46** and the third connecting portion **47** are provided with corresponding slots **48** for holding the balls **3**.
[0034] The three rows of the balls **3** comprises a first ball row **31** and a second ball row **32** contacting an interior side of the long wall portion **14**, and a third ball row **33** contacting an interior side of the short wall portion **15** and located between the first ball row **31** and the second ball row **32** in a vertical direction.
[0035] Referring to FIGS. **1-8**, the inner slide **2** is a roll formed steel plate of uniform thickness. The inner slide **2** comprises a first inner slide end **21** housed in the first outer slide **11**, a second inner slide end **22** housed in the second outer slide **12**, and a connecting wall **23** between the first inner slide end **21** and second inner slide end **22**. The connecting wall **23** is tilted at a certain angle to reasonably configure the long wall part **14** and the short wall part **15** of the first outer slide **11** and second outer slide **12** on each side in a staggered form, thus reducing the size of the ball bearing slide **100** in the height direction as much as possible. The first inner slide end **21** and the second inner slide end **22** have the same structure and are arranged in a central symmetry form. The second inner slide end **22**, for example, comprises a first curved part **24** against a curved surface of the first ball row **31** facing the inner slide **2**, a second curved part **25** against a curved surface of the second ball row **32** facing the inner slide **2**, and a third curved part **26** against a curved surface of the third ball row **33** facing the inner slide **2**. The third curved **26** and the short wall portion **15** hold

the third ball row **33** from both sides thereof in the width direction. In the vertical direction, the third curved part **26** is located on a perpendicular line D of a line connecting centers of the first ball row **31** and the second ball row **32**. According to a preferred embodiment, the third curved part **26** is symmetrical with respect to the perpendicular line D (see FIG. **9**). A center point of the inner slide **2** is d (see FIG. **7**), and an axis extending in the vertical direction through the point d is defined as a central axis C-C. The third curved part **26** has a bottom side **263** and an interior side **264** in the width direction, wherein the bottom side **263** exceeds the central axis C-C of the ball bearing slide **100** in a thickness direction. A contour of the interior side **264** of the third curved part **26** has an angle b. Referring to FIG. **9**, point A is a center of circle of the third ball row **33**, point B is a center of circle of the interior side **264** of the third curved part **26**, and point C is a contact point between the third ball row **33** and the interior side **264**. The points A, B and C are located in the same straight line, and the angle b is a curved angle of the third curved part **26**. The angle b ranges from 100 degrees to 110 degrees and can be 100 degrees or 110 degrees. According to the preferred embodiment, the angle b is 110 degrees, while in other embodiments, the angle can be more than 110 degrees and less than 180 degrees. Here the angle is an obtuse angle. When a top surface of the second outer slide **12** receives an external pressure, the third ball row **33** can transmit the pressure to the second inner slide end **22**, and the third curved part **26** can bear the pressure in the vertical direction on an obtuse angle contact surface, so as to reasonably and uniformly distribute the bearing forces of the third ball row **33** and the inner slide **2**. A tail portion **261** of the third curved part **26** is cocked towards the third ball row **33**, and an interior side of the tail portion **261** is maintained with a curved profile. A tangent line of a center point of the third ball row **33** at an end of the tail portion **261** of the third curved part **26** and a tangent line of the short wall portion **15** together form an opening **262** with a first angle a. The first angle a can be adjusted according to a thickness of the ball cage **4** to allow the ball cage **4** to extend from an opening range of the angle a, which is conducive to the effective support of the ball cage **4** and the overall stability of the ball cage **4**.

[0036] The inner slide **2** further comprises a first convex part **27** connecting the first curved part **24** and the second curved part **25**, which corresponds to a position of the third curved part **26**; and a second convex part **28** connecting the second curved part **25** and the third curved part **26**. The first convex part **27** is formed by folding and bending a portion between the first curved portion **24** and the second curved portion **25**. The second convex part **28** is formed by folding and bending a portion between the first curved part **24** and the third curved part **26** and rolling the part at an angle towards the second holding portion **42**. The bottom side **263** of the third curved part **26** is pressed inwardly towards the first convex part **27**, so that the bottom side **263** of the third curved part **26** exceeds the other side of the central axis C, thereby compressing the thickness of the ball bearing slide as much as possible and making the long wall portion **14** as compact as possible. The first convex part **27** is located between the first ball row **31** and the second ball row **32**; the first convex part **27** is spaced from the first connecting part **45** in a width direction with a first spacing **451** therebetween. One side of the first convex part **27**, which faces the first connecting portion **45**, is a flat portion **271**, wherein the flat portion **271** is rolled into a flat surface to place the ball cage **4**. The second convex part **28** is tilted towards the second holding portion **42** in the vertical direction, and can bear forces in both the vertical direction and the width direction according to a force direction of the outer slide **1**. The second convex part **28** is located between the first ball row **31** and the third ball row **33**. A thickness of the second convex part **28** equals a distance between the first ball row **31** and the third ball row **33**; and dimension of the second convex part **28** is smaller than a total thickness of two rolled raw materials of the inner slide **2**. Because the inner slide **2** squeezes a thickness of the second convex part **28** during roll-forming, making the raw material elongated, which in turn allows the spacing between the first ball row **31** and the third ball row **33** to be smaller than the total thickness of two materials. As a result, the size of the ball bearing slide **100** is comprehensively reduced in the thickness direction. The second convex part **28** is spaced

from the second holding portion **42** with a second spacing **422** therebetween. The second holding portion **42** has an internal wall **421** which faces the second convex part **28**, and the internal wall **421** is a flat surface. One side of the second convex part **28**, which faces the internal wall **421**, is rolled into a flat surface **281** to place the ball cage **4**. When rolling the inner slide **2**, restriction mold surfaces are added to the flat portion **271** of the first convex part **27** and the flat surface **281** of the second convex part **28**, and thus avoid material flow excessive extension of the convex parts **27** and **28** caused by excessive extrusion during rolling. The material flow excessive extension will reduce the design space of the ball cage **4**, resulting in the need of corresponding transition part for the ball cage **4**, which in turn makes the ball cage **4** complex, and increases the size of the ball bearing slide **100** in the overall thickness direction.
[0037] The connecting wall **23** of the inner slide **2** is tilted because the third curved part **26** in the first outer slide **31** and the third curved part **26** in the second outer slide **32** are set back-to-back. Therefore, the long wall portion **14** should be correspondingly arranged at the side with the third curved part **26**, and the short wall portion **15** should be arranged at the opposite position. In order to compress the overall height of the ball bearing slide as much as possible, the long wall portion **14** should cooperate with the short wall portion **15**.
[0038] The inner slide **2** of the present invention is designed to have the first curved part **24**, the second curved part **25** and the third curved part **26** at corresponding positions, as well as the first convex part **27** and the second convex part **28** located between the ball rows **31**, **32** and **33**. A radian of the third curved part **26** should be obtuse curved surface as far as possible, and the third curved part **26** should be symmetrically arranged relative to the symmetry axis of the line connecting the centers of the ball rows **31** and **32**, so that forces along the height direction of the ball bearing slide **100** can be uniformly received in the curved surface, which improves the overall bearing capacity of the ball bearing slide **100**. Both the first convex part **27** and the second convex part **28** have flat surfaces to make as much room as possible for the design space of the ball cage **4**, which effectively compresses the overall height and width of the ball bearing slide **100**. The third curved part **26** is set along the symmetry axis of the line connecting the centers of the ball rows **31** and **32**, and towards them as close as possible, so that the thickness of the second convex part **28** located between the first ball row **31** and the third ball row **33** is rolled and compressed as much as possible, and the spacing therebetween is less than the total thickness of two material thicknesses of the inner slide **2**.
[0039] With the foregoing design, the overall thickness of the ball bearing slide **100** is directly reduced, so that the design of the inner slide **2** can fully save the space between the outer slides **1**, thus leaving enough room for the ball cage **4** and making full use of the space within the outer slide **1**. In general, sufficient internal space of the ball cage **4** is ensured, which rationalizes the ball cage arrangement and improves the overall bearing capacity of the ball bearing slide.
[0040] Finally, it should be noted that the above embodiments are used only to illustrate the technical solution of the present invention and not to limit it. Despite the detailed description of the present invention with reference to the foregoing embodiments, it should be understood by those of ordinary skill in the art that it is still possible to modify the technical solution recorded in the foregoing embodiments or to make equivalent substitutions for part or all of the technical features thereof. Such modifications or substitutions do not make the essence of the corresponding technical solution different from those in the embodiments of the present invention.
[0041] The above description is only part of all embodiments of the present invention. Any equivalent changes to the technical solution of the present invention made by those skilled in the art after reading the specification are covered by the following claims.

Claims

1. A ball bearing slide, comprising: at least one outer slide (**1**), an inner slide (**2**) located within the outer slide (**1**), a ball cage (**4**) located between the inner slide (**2**) and the outer slide (**1**), and three rows of balls (**3**) rollably arranged in the ball cage (**4**); wherein the inner slide (**2**) is supported by the balls (**3**) to slide relative to the outer slide (**1**) along an arrangement direction of the three rows of the balls (**3**); the outer slide (**1**) comprises a bottom wall (**13**), as well as a long wall portion (**14**) and a short wall portion (**15**) which are formed by integrally extending both sides of the bottom wall (**13**) and are arranged along a length direction of the bottom wall (**13**); the three rows of the balls (**3**) comprises a first ball row (**31**) and a second ball row (**32**) contacting an interior side of the long wall portion (**14**), and a third ball row (**33**) contacting an interior side of the short wall portion (**15**) and located between the first ball row (**31**) and the second ball row (**32**) in a vertical direction; wherein the inner slide (**2**) comprises a first curved part (**24**) against a curved surface of the first ball row (**31**) facing the inner slide (**2**), a second curved part (**25**) against a curved surface of the second ball row (**32**) facing the inner slide (**2**), and a third curved part (**26**) against a curved surface of the third ball row (**33**) facing the inner slide (**2**); the third curved part (**26**) is located between the first ball row (**31**) and the second ball row (**32**) in the vertical direction, and is symmetrically arranged relative to a symmetry axis of a line connecting centers of the first ball row (**31**) and the second ball row (**32**).
2. The ball bearing slide, as recited in claim 1, wherein the third curved part (**26**) has a bottom side (**263**) in a width direction perpendicular to the vertical direction; the inner slide (**2**) has a central axis C-C extending in the vertical direction and passing through a central point of the inner slide (**2**); and the bottom side (**263**) exceeds the central axis C-C in the width direction.
3. The ball bearing slide, as recited in claim 2, wherein the ball cage (**4**) comprises a fourth holding portion (**44**) corresponding to the short wall portion (**15**) for holding the third ball row (**33**), wherein the fourth holding portion (**44**) extends flatly along the vertical direction.
4. The ball bearing slide, as recited in claim 3, wherein the ball cage (**4**) further comprises a first holding portion (**41**) for holding the first ball row (**31**), a third holding portion (**43**) for holding the second ball row (**32**), and a first connecting portion (**45**) connecting the first holding portion (**41**) and the third holding portion (**43**); wherein the first connecting portion (**45**) extends flatly along the vertical direction, and a portion of the long wall portion (**14**) corresponding to the first connecting portion (**45**) also extends flatly along the vertical direction.
5. The ball bearing slide, as recited in claim 4, wherein the outer slide (**1**) further comprises: a first support (**141**) formed at an end of the long wall portion (**14**) and having a curved internal contour, a second support (**151**) formed at an end of the short wall portion (**15**) and having a curved internal contour, a first corner (**16**) corresponding to the first support (**141**) and vertically arranged at a top of the long wall portion (**14**), a second corner (**17**) corresponding to the second support (**151**) and vertically arranged at a top of the short wall portion (**15**), and a connecting wall (**18**) connecting the first corner (**16**) and the second corner (**17**); wherein the first holding portion (**41**) of the ball cage (**4**) is arranged corresponding to the first corner (**16**), a second holding portion (**42**) of the ball cage (**4**) is arranged corresponding to the second corner (**17**), the third holding portion (**43**) is arranged corresponding to the first support (**141**), and the fourth holding portion (**44**) is arranged corresponding to the second support (**151**); the first ball row (**31**) is restricted by the first holding portion (**41**) corresponding to the first corner portion (**16**), the second ball row (**32**) is restricted by the third holding part (**43**) corresponding to the first support (**141**), and the third ball row (**33**) is restricted by the fourth holding part (**44**) corresponding to the second support (**151**).
6. The ball bearing slide, as recited in claim 1, wherein an angle of an internal contour of the third curved part (**26**) is no less than 100° while no more than 110° for holding the third ball row (**33**).
7. The ball bearing slide, as recited in claim 5, wherein the inner slide (**2**) further comprises a second convex part (**28**) connecting the second curved part (**25**) and the third curved part (**26**), wherein the second convex part (**28**) is formed by folding and bending a portion between the first

curved part (**24**) and the third curved part (**26**) and rolling the part at an angle towards the second holding portion (**42**); the second convex part (**28**) is located between the first ball row (**31**) and the third ball row (**33**).

8. The ball bearing slide, as recited in claim 7, wherein a thickness of the second convex part (**28**) equals a distance between the first ball row (**31**) and the third ball row (**33**), a dimension of the second convex part (**28**) is smaller than a total thickness of two rolled raw materials of the inner slide (**2**).

9. The ball bearing slide, as recited in claim 6, wherein the second convex part (**28**) is spaced from the second holding portion (**42**) with a second spacing (**422**) therebetween; one side of the second convex part (**28**), which faces the second holding portion (**42**), is rolled into a flat surface to place the ball cage (**4**).

10. The ball bearing slide, as recited in claim 7, wherein the inner slide (**2**) further comprises a first convex part (**27**) connecting the first curved part (**24**) and the second curved part (**25**), which corresponds to a position of the third curved part (**26**), wherein the bottom side (**263**) of the third curved part (**26**) is pressed inwardly towards the first convex part (**27**); the first convex part (**27**) is formed by folding and bending a portion between the first curved portion (**24**) and the second curved portion (**25**).

11. The ball bearing slide, as recited in claim 8, wherein the first convex part (**27**) is located between the first ball row (**31**) and the second ball row (**32**); the first convex part (**27**) is spaced from the first connecting part (**45**) in a width direction with a first spacing (**451**) therebetween; one side of the first convex part (**27**), which faces the first connecting portion (**45**), is rolled into a flat surface to place the ball cage (**4**).

12. The ball bearing slide, as recited in claim 1, wherein a tail portion (**261**) of the third curved part (**26**) is cocked towards the third ball row (**33**); a tangent line of the tail portion (**261**) of the third curved part (**26**) and a tangent line of the short wall portion (**15**) together form an opening (**262**) with a first angle (a); an end of the fourth holding portion (**44**) extends out through the opening (**262**).